PHANIDHAR AKULA

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AI & Full-Stack Software Engineering • Production AI & Backend Systems • U.S. work authorization: OPT pending for Oct 2026; STEM OPT eligible

EDUCATION

Miami University | M.S. in Computer Science — *Aug 2024 – Aug 2026*
Oxford, OH • GPA: 3.86/4.00 • Full Scholarship • Assistantship
Thesis: SimForge – A Reproducible Cross-Simulator Benchmarking Framework for Urban Traffic Simulation (defended, July 2026)
Coursework: Generative AI, Machine Learning, Advanced Database Systems, Software Quality Assurance, Cryptography

Malla Reddy University | B.Tech in Computer Science — *Aug 2020 – May 2024*
Hyderabad, India • CGPA: 8.2/10 • Two peer-reviewed publications in CNN-based soil classification and LDA topic modeling

EXPERIENCE

Miami University | Graduate Assistant – Department of CSE — *Aug 2025 – May 2026*
Research group of Dr. DJ Rao
- Graduate research in the simulation/digital-twin group; lead developer on Cityscape, integrated into the SimForge thesis pipeline.
- Co-administered the Grand Challenges Scholars Program (GCSP), mentoring undergraduate research while supporting faculty-led research initiatives and program operations.

Smart Pathfinders Overseas Education | Software Engineer Intern — *May 2023 – Jul 2024*
- Built an internal staff management and organizational hierarchy platform for an early-stage startup.
- Shipped production React/TypeScript features, collaborating with product and design on iterative releases while resolving UI regressions and edge cases identified through post-release feedback.

Independent Clients | Freelance Software Engineer — *Aug 2023 – May 2024*
- Delivered end-to-end full-stack features for three early-stage startup clients under NDA.
- Built React/TypeScript frontends and Python/MySQL backend services exposed through REST APIs, maintaining AWS infrastructure while balancing performance, security, and maintainability.

SELECTED PROJECTS

LumiAI | Production AI Tutoring Platform – studywithlumi.com — *Nov 2025 – Present*
React, TypeScript, LLMs, Supabase (PostgreSQL, Auth, Storage), Vercel Serverless, OAuth
- Solo-architected, built, deployed, and now maintain a production AI tutoring platform serving 60+ active students with document-grounded AI chat, quiz and flashcard generation, SM-2 spaced repetition, and voice mode; currently building Android and iOS companion apps.
- Built the LLM serverless layer with token-verified requests, server-side API-key isolation, anonymous-call rejection, document-grounded context from user-uploaded materials, streaming responses, and prompt caching.
- Owned the full stack across React/TypeScript, PostgreSQL row-level security, Google OAuth, hardened SECURITY DEFINER functions, admin analytics, storage workflows, and production deployment on Vercel.

Cityscape | Open-Source Population Synthesis and Geospatial Demand Generation — *Aug 2025 – Jun 2026*
C++, OpenMP, OSM/Geofabrik, US Census PUMS, OSC HPC (Pitzer)
- Authored three merged C++ pull requests (github.com/raodj/cityscape, PRs #1–#3) to an open-source population-synthesis and trip-demand engine, fixing OSM building-classification and data-integrity logic across five metros.
- Removed ~1.25M stranded-population artifacts, collapsed LA synthetic-home ratios from ~70% to ~5%, and ran a 4-city, 16-combination HPC sweep that raised worst-case calibration R^2 from 0.812 to 0.917.

SimForge | Master's Thesis – A Reproducible Cross-Simulator Benchmarking Framework for Urban Traffic Simulation — *Aug 2024 – Jul 2026*
Python, OpenMP, multiprocessing, Docker, Apptainer, SLURM, SUMO, MATSim, DTALite, OSM, US Census PUMS
- Built an open-source benchmarking framework integrating SUMO, MATSim, and DTALite under a canonical dataset pipeline for fair cross-simulator evaluation; released under Apache-2.0 with a citable Zenodo DOI.
- Reduced shared-route pre-routing from 141.9 hours to 7.1 hours through deterministic route caching and 16-way parallelism; benchmarked three cities up to 500K trips and found up to 96% travel-time divergence under saturation.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL, Java
AI/ML & Data: LLMs, RAG-style document grounding, prompt engineering, PyTorch, TensorFlow, CNNs, NLP, inference, US Census PUMS pipelines
Full-Stack & Backend: React, Node.js, Vercel Serverless, FastAPI, REST APIs, PostgreSQL, MySQL, Supabase Auth/Storage, OAuth, row-level security
Cloud & Systems: AWS, Docker, GitHub Actions, CI/CD, Linux, Git, OpenMP, multiprocessing, SLURM, Apptainer
Engineering & Simulation: pytest, mutation testing, unit testing, code review, API-key isolation, reproducibility guards, performance optimization, OSM/Geofabrik, digital twins, agent-based traffic simulation

PUBLICATIONS

- **Akula, P.**, & Rao, D. M. Attributable Cross-Simulator Benchmarking for Urban Traffic Simulation: Reproducibility, Consistency, and Regime-Dependent Divergence. IEEE Transactions on Intelligent Transportation Systems (under review, first author, 2026).
- **Akula, P.**, Stojanovski, S., & Rao, D. M. CITYSCAPE: A Digital Twin Framework for Temporospatial Modeling of Work-Commute Dynamics and Roadway Infrastructure Demand. Frontiers (under review, first author, 2026).

LEADERSHIP

- President, Graduate Students of Color Association (104 members; Certificate of Appreciation, 2026), and elected Board Member, Graduate Student Council, Miami University (Aug 2025 – May 2026). Led recruitment, events, and graduate student advocacy.